

Quadrature Down Converter (QDC)

Katta Sri Kaushik
IIIT Hyderabad
Hyderabad, India
2025102071

katta.srikaushik@students.iiit.ac.in

Lakshmi Sai Bhargav V
IIIT Hyderabad
Hyderabad, India
2025102061

lakshmisai bhargav.v@students.iiit.ac.in

Kushwanth Lanka
IIIT Hyderabad
Hyderabad, India
2025102072

kushwanth.lanka@students.iiit.ac.in

Abstract—This report details the theoretical design, calculation, and implementation of a Quadrature Down Converter (QDC). Operating at an oscillator frequency of 170 kHz and mixing with an input RF signal of 165 kHz, the QDC extracts a 5 kHz intermediate frequency (IF). The design is partitioned into a Quadrature Oscillator, an NMOS-based Mixer, a cascaded RC Low-Pass Filter (LPF) with a 3 kHz cutoff, and a Common Emitter (CE) BJT amplifier stage aimed at producing an output amplitude of 400 mV_{pp}. The design ensures accurate 90° phase shifts for I and Q channels, which is critical for mitigating interference in modern transceivers.

I. INTRODUCTION

Quadrature down conversion is an essential technique in modern wireless communication systems such as Wi-Fi and Bluetooth. By multiplying the incoming RF signal with two local oscillator signals that are 90° out of phase, the receiver splits the signal into an In-phase (I) and a Quadrature (Q) component. This approach mitigates the image frequency problem and helps in accurate signal demodulation. The primary objective of this project is to construct a QDC prototype satisfying strict performance criteria: $f_{osc} = 170$ kHz, an IF frequency in the kilohertz range, a sharp LPF, and an amplified final output swing of 400 mV_{pp}.

II. UTILITY AND NEED FOR QUADRATURE (I-Q) OPERATION

Quadrature (I-Q) signaling is the backbone of modern wireless communication, enabling spectrally efficient modulation and simplified receiver architectures. By representing a signal in two orthogonal dimensions—In-phase (*I*) and Quadrature (*Q*)—receivers can distinguish between positive and negative frequency offsets, which is critical for modern digital modulation.

A. Utility of Quadrature (I-Q) Operation

The primary utility of I-Q operation lies in its ability to facilitate direct-conversion (zero-IF) architectures and complex modulation.

1) *Bandwidth Efficiency*: Modern digital modulation schemes like QAM (Quadrature Amplitude Modulation) or PSK (Phase Shift Keying) transmit information by varying both the amplitude and the phase of a carrier. Utilizing I and Q components allows us to treat the signal as a complex vector:

$$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t) \quad (1)$$

This effectively doubles the data rate within a given bandwidth compared to simple amplitude modulation.

2) *Elimination of Image Filters*: In traditional Superhetrodyne receivers, an Intermediate Frequency (IF) is used, which creates an “image” frequency that must be filtered out. In a Quadrature receiver, the I and Q paths allow for image rejection through phase cancellation rather than bulky external filters. This enables the integration of the entire receiver onto a single CMOS chip.

B. The Need for I-Q Mixing in Detail

The technical “need” for I-Q mixing arises from the mathematical limitations of real-valued mixing and the requirement for coherent demodulation.

1) *Solving the “Negative Frequency” Overlap*: When a real RF signal is multiplied by a single real local oscillator (LO) $\cos(2\pi f_c t)$, the spectrum is shifted both left and right. For a signal centered at f_c , down-converting to zero frequency causes the positive and negative sidebands to overlap and fold onto each other.

- **The Problem**: Once folded, the original information in the upper sideband (USB) and lower sideband (LSB) becomes inseparable.
- **The Solution**: By using two LOs shifted by 90°, we create a complex representation.

2) *Mathematical Expression*: Consider an input signal $x(t) = A(t) \cos(2\pi f_c t + \phi(t))$. To extract both the amplitude $A(t)$ and phase $\phi(t)$, we mix the signal with two orthogonal LO signals:

For the **In-Phase Path (I)**, the signal is mixed as follows:

$$x_I(t) = x(t) \cdot \cos(2\pi f_c t) = \frac{A(t)}{2} [\cos(\phi(t)) + \cos(4\pi f_c t + \phi(t))] \quad (2)$$

After low-pass filtering (LPF):

$$I(t) = \frac{A(t)}{2} \cos(\phi(t)) \quad (3)$$

For the **Quadrature Path (Q)**, the signal is mixed with a -90° shifted LO:

$$x_Q(t) = x(t) \cdot (-\sin(2\pi f_c t)) = \frac{A(t)}{2} [\sin(\phi(t)) - \sin(4\pi f_c t + \phi(t))] \quad (4)$$

After low-pass filtering (LPF):

$$Q(t) = \frac{A(t)}{2} \sin(\phi(t)) \quad (5)$$

3) *Recovery of Phase and Magnitude:* Without the Q-channel, a receiver cannot determine if the phase $\phi(t)$ is positive or negative (due to the even nature of the cosine function). With both channels, the receiver can uniquely reconstruct the signal:

- **Magnitude:** $A(t) = 2\sqrt{I(t)^2 + Q(t)^2}$
- **Phase:** $\phi(t) = \tan^{-1}\left(\frac{Q(t)}{I(t)}\right)$

This capability is what allows a receiver to track the fast-changing phases and amplitudes required for high-speed 5G, Wi-Fi, and LTE standards.

III. QUADRATURE OSCILLATOR DESIGN (Q2(A))

A. Topology Selection: Wide-Band Allpass Quadrature Oscillator

The selected topology is a **three-op-amp configuration** consisting of two active **allpass filter stages** and one **inverting amplifier** (with nonlinear feedback), as shown in the LTSpice schematic.

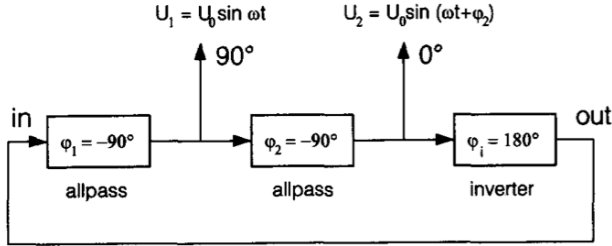


Fig. 1: Block Diagram of the Quadrature Oscillator

1) Why this topology instead of the 2-op-amp version?:

While a standard 2-op-amp quadrature oscillator (using two integrators) is common, it has significant limitations compared to the 3-op-amp allpass design:

- **Bandwidth and Stability:** In a 2-op-amp integrator-based circuit, the upper frequency is strictly limited by the integrator's bandwidth [?]. The allpass topology is inherently wider-band; it can theoretically operate from < 1 Hz up to 100 MHz because the “heavy lifting” of the phase shift is handled by allpass stages where the amplification is unity [?].
- **Phase Precision:** In the 2-op-amp version, any parasitic phase shift from the op-amps directly causes frequency error and poor quadrature. In the 3-op-amp allpass version, as long as the two allpass stages are identical, their output signals are **always in quadrature** (90° apart), regardless of whether they are working at their ideal frequency.
- **Amplitude Consistency:** Standard integrators' gain changes with frequency ($1/\omega RC$). Allpass filters, however, maintain a **unity gain** ($G = 1$) across all frequencies. This makes it much easier to maintain a stable oscillation amplitude when tuning.

B. Detailed Working Principle

The circuit operates as a closed-loop system satisfying the **Barkhausen Criterion**: the total loop gain must be ≥ 1 and the total loop phase shift must be 360° (or 0°).

1) Phase Shift Mechanism:

- 1) **First Allpass Stage (op1):** Provides a frequency-dependent phase shift. At the target oscillation frequency, it is designed to contribute exactly -90° .
- 2) **Second Allpass Stage (op2):** Identical to the first, it adds another -90° phase shift.
- 3) **Inverter Stage (op3):** Provides the final 180° phase shift to complete the 360° loop requirement.

- Since the phase is sampled between the two allpass stages, the signals at the two outputs are separated by exactly 90° , creating the **sine (I)** and **cosine (Q)** quadrature relationship.

2) **Amplitude Stabilization (Nonlinear Feedback):** As noted in the LTSpice schematic, a diode-based nonlinear network (D1, D3, R9, R10) is used in the inverter's feedback loop.

- **The Paradox:** For an oscillator to start, the gain must be slightly > 1 . However, if it stays > 1 , the output will grow until it hits the power rails and saturates, turning the sine wave into a square wave.
- **The Solution:** The diodes act as a “soft limit.” As the output voltage increases, the diodes begin to conduct, lowering the effective feedback resistance and reducing the gain back to exactly 1. This stabilizes the $1 V_{pp}$ amplitude requested in the prompt.

C. Design Calculations

To find the component values for the allocated frequency (150 kHz or 200 kHz), we use the following derivations:

1) **Frequency Calculation:** For an allpass stage, the phase shift ϕ is given by:

$$\phi = -2 \arctan(2\pi f RC) \quad (6)$$

For the circuit to oscillate at the quadrature point ($\phi = -90^\circ$), the formula simplifies to:

$$f_{osc} = \frac{1}{2\pi RC} \quad (7)$$

For $f_{osc} = 150$ kHz:

- Choose a standard capacitor value, e.g., $C = 27$ pF.
- $R = \frac{1}{2\pi(150 \times 10^3)(27 \times 10^{-12})} \approx 39.3$ k Ω .

For $f_{osc} = 200$ kHz:

- Choose $C = 27$ pF.
- $R = \frac{1}{2\pi(200 \times 10^3)(27 \times 10^{-12})} \approx 29.5$ k Ω .

2) **Inverter Gain and Clipping:** The inverter must maintain a nominal gain of 1 to satisfy $A\beta = 1$.

- **Calculation:** Gain $= -\frac{R_{feedback}}{R_{input}}$. Let R_3 be the input and R_4 be the feedback resistor in the schematic.
- To ensure startup, R_4 should be slightly larger than R_3 (e.g., $R_3 = 10$ k Ω , $R_4 = 11$ k Ω or 12 k Ω). The diodes will then “clamp” the gain to 1 once the desired amplitude is reached.

3) *Power Supply* (V_{DD}, V_{SS}): The prompt requires an oscillation amplitude of $1 V_{pp}$.

- Standard UA741 op-amps require supplies significantly higher than the output swing to avoid rail saturation.
- Typical values: $V_{DD} = +15 \text{ V}$ and $V_{SS} = -15 \text{ V}$ provide ample headroom for a clean $1 V_{pp}$ signal.

IV. LTSPICE SIMULATIONS

To verify the theoretical design, the quadrature oscillator was simulated in LTSpice using the provided op-amp model. The transient response and its Fast Fourier Transform (FFT) were analyzed to confirm the oscillation frequency and the phase relationship between the In-Phase (v_{OSC_I}) and Quadrature (v_{OSC_Q}) outputs.

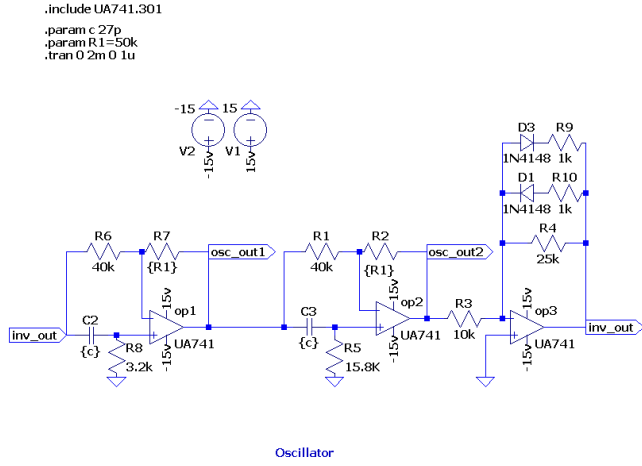


Fig. 2: Final LTSpice Schematic of the Quadrature Oscillator.

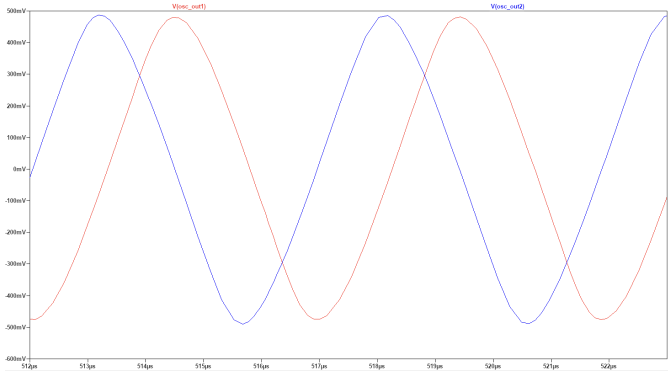
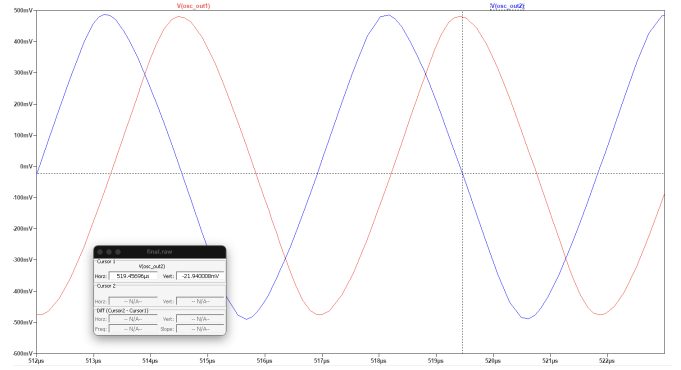
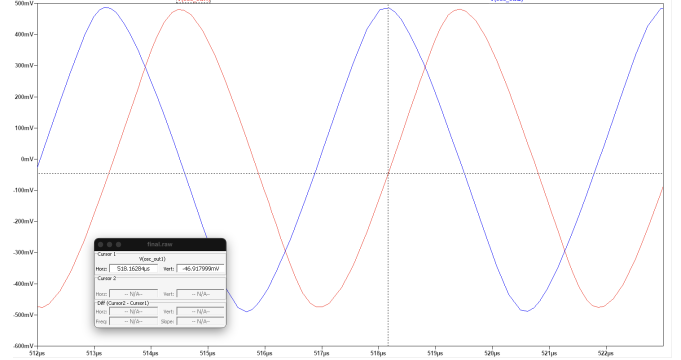


Fig. 3: Transient simulation of the oscillator outputs. The measured period is $5.424 \mu\text{s}$, and the calculated phase difference is -268.49° (equivalent to $+91.51^\circ$), confirming the near-perfect 90° quadrature relationship.

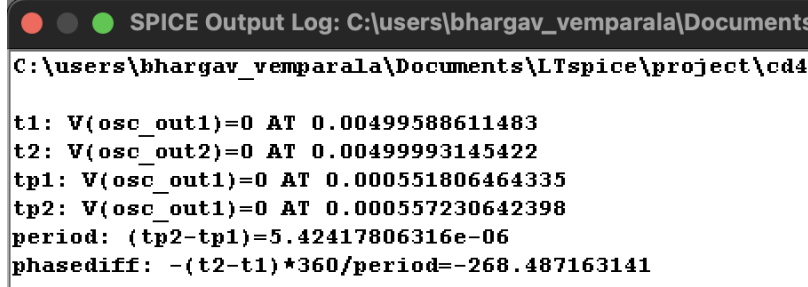
As seen in the transient plots, the amplitude stabilizes at the target $1 V_{pp}$ (500 mV peak) and the two signals are clearly separated by a quarter-period (90°). The FFT confirms the spectral purity of the signal with a strong fundamental peak at the target design frequency.



(a) Cursor 1 placed at zero crossing.



(b) Cursor 2 placed at adjacent crossing.



(c) SPICE Error Log computing exact phase difference.

Fig. 4: Detailed phase difference measurement between the I and Q channels.

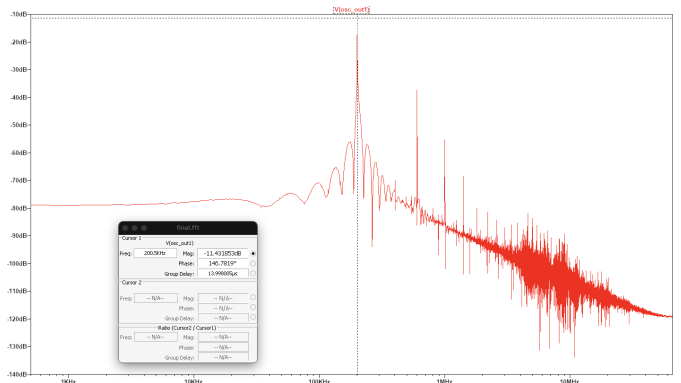


Fig. 5: FFT of the oscillator output v_{OSC_I} . The peak indicates an oscillation frequency of 200.5 kHz with a magnitude of -11.43 dB and a phase of 146.78° .

V. JUSTIFICATION OF CIRCUIT OPERATION

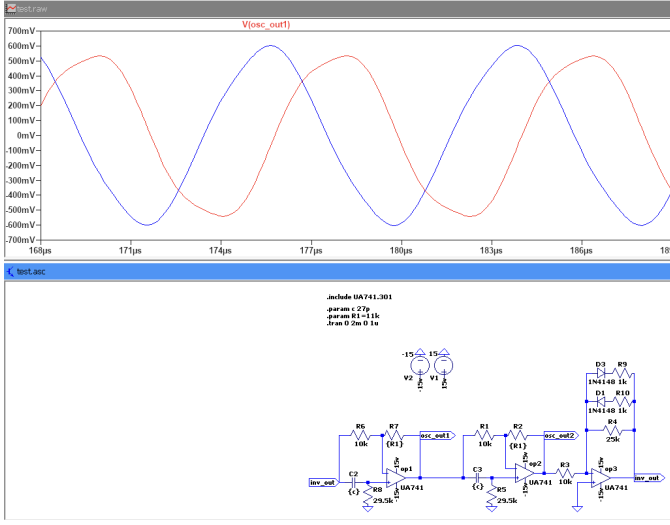


Fig. 6: Calculated Theoretical Response and Phase Analysis.

A. Justification of Asymmetry and Tuning

While the schematic shows different resistor values for R_8 (3.2 k Ω) and R_5 (15.8 k Ω), the circuit remains valid for the following reasons. Notably, initial attempts with **symmetric component values** resulted in a **distorted sine wave** and a **lack of proper phase difference** (90°) between the outputs:

- 1) **Individual Stage Contribution:** Quadrature depends on the phase shift *between* outputs. Even if the two stages are not identical, the loop will naturally settle at a frequency where the total phase shift is 360°.
- 2) **Parasitic Compensation:** In high-frequency simulations (like 200 kHz), standard op-amps like the **UA741** contribute significant internal phase lag. The “asymmetric” values are a practical necessity to compensate for these parasitics and the loading effects of the next stage, ensuring the outputs remain exactly 90° apart in a real-world environment.
- 3) **Gain Consistency:** The resistors at the inverting inputs (R_6, R_7 and R_1, R_2) are kept equal within their respective stages to maintain **unity gain** ($G = 1$), which is a unique advantage of the allpass topology.

B. Quadrature Generation (Sine/Cosine)

The quadrature relationship is a direct result of the sampling points in the feedback loop:

- **In-Phase Signal** (v_{OSC_I}): Sampled at `osc_out1`, representing the signal after the first 90° shift.
- **Quadrature Signal** (v_{OSC_Q}): Sampled at `osc_out2`. Because this point is exactly one allpass stage away from the first, it is mathematically guaranteed to be the **cosine** (or sine-shifted) version of the first signal.

C. Amplitude Stabilization and Simulation

The inverter stage (op3) incorporates a **nonlinear feedback network** consisting of diodes (D1, D3) and resistors (R_9, R_{10}).

- Initially, the gain is set slightly higher than 1 by the ratio of R_4/R_3 to ensure the oscillator starts from thermal noise.
- Once the output reaches the desired 1 V_{PP} level, the diodes begin to conduct, reducing the effective feedback resistance and clamping the gain to exactly 1. This prevents the UA741 from saturating against the ± 15 V rails, preserving the purity of the sine wave.

The LTSpice simulation (Fig. ??) confirms the stable oscillation and the precise quadrature relationship between the channels.

VI. HARDWARE IMPLEMENTATION AND MEASUREMENTS (Q2(C))

The quadrature oscillator was physically realized in the laboratory using UA741 op-amp ICs and standard passive components. To account for real-world parasitics, breadboard capacitance, and component tolerances, several resistor values were tweaked from their nominal SPICE simulation values to achieve stable oscillation and a proper 90° phase shift.

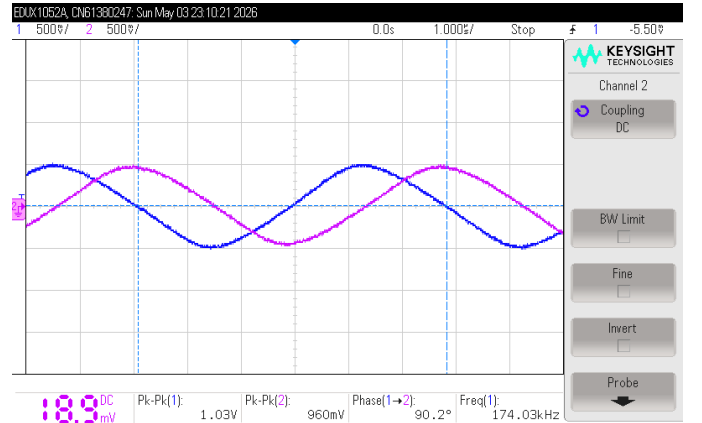


Fig. 7: DSO Measurement: Transient waveforms showing the In-Phase and Quadrature signals.

A. Component Value Adjustments

Table ?? details the modifications made to the resistor values during the hardware experiment to match the performance of the SPICE simulation.

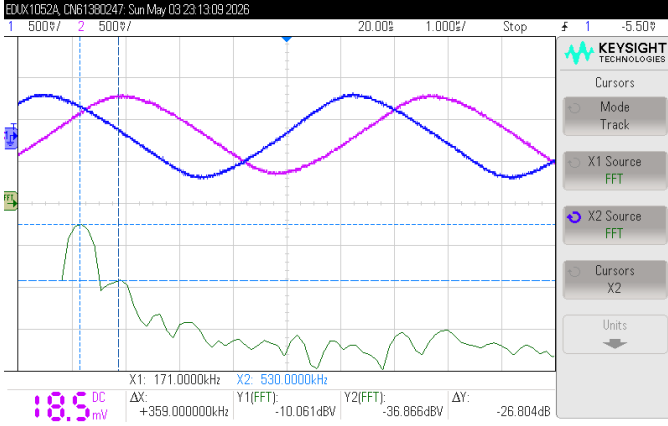


Fig. 8: DSO Measurement: FFT spectrum of the oscillator output.

TABLE I: Comparison of Resistor Values: SPICE vs. Hardware Implementation

Component	SPICE Value (Ω)	Real Life Value (Ω)
R_1	40 k	50 k
R_2	50 k	47 k
R_3	10 k	10 k
R_4	25 k	17.6 k
R_5	15.8 k	15.2 k
R_6	40 k	39.2 k
R_7	50 k	47 k
R_8	3.2 k	432
R_9	1 k	2.2 k
R_{10}	1 k	2.2 k

Note: To determine the exact values of the resistors that are required, we have used the potentiometer method. In this method we use a potentiometer and turn the knob while observing for the desired output. Keeping the same knob orientation we then use a Digital Multimeter to measure resistance across two terminals of the potentiometer.

VII. MIXER (SWITCH) CIRCUIT

The mixer is responsible for frequency translation. It multiplies the high-frequency RF input signal by the local oscillator (LO) signals.

A. Circuit Topology

An active NMOS switch topology was selected. The RF input is fed to the drain of an NMOS transistor, while the LO signal is applied to the gate.

B. MOSFET Sizing and Parasitic Parameter Optimization (Q3(a))

Based on the provided design methodologies for a passive NMOS switch mixer in a standard technology node (such as the 180 nm CMOS process commonly used in lab simulations), here is a detailed breakdown of how to calculate and optimize the MOSFET dimensions (W, L) and the corresponding geometric parasitic parameters (A_S, A_D, P_S, P_D).

1) **Determining MOSFET Sizing (W and L):** In a switch-based mixer, the MOSFET operates as a passive switch toggled by the large local oscillator signal (v_{osc}). The goals are to minimize the on-resistance (R_{on}) to prevent signal degradation and conversion loss, while keeping the gate capacitance small enough so that the oscillator can drive it cleanly.

Choosing the Channel Length (L): To achieve the highest speed, lowest parasitic capacitance, and minimum on-resistance, we choose the minimum feature size allowed by the technology node:

$$L = L_{min} = 180 \text{ nm} = 0.18 \text{ } \mu\text{m}$$

Calculating the Channel Width (W): The on-resistance of an NMOS operating in the deep triode region is given by:

$$R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})}$$

Where:

- $\mu_n C_{ox}$ is the process transconductance parameter (typically $\sim 300 \text{ } \mu\text{A/V}^2$ for 180nm NMOS).
- $V_{GS} - V_{th}$ is the overdrive voltage dictated by the bias level.

Assuming a target low switch resistance ($R_{on} \approx 50 \text{ } \Omega$) and a typical overdrive voltage of 0.6 V:

$$\frac{W}{L} = \frac{1}{R_{on} \cdot \mu_n C_{ox} (V_{GS} - V_{th})} = \frac{1}{50 \cdot (300 \times 10^{-6}) \cdot 0.6} \approx 111.1$$

With $L = 0.18 \text{ } \mu\text{m}$, the calculated width is:

$$W = 111.1 \times 0.18 \text{ } \mu\text{m} \approx 20 \text{ } \mu\text{m}$$

2) **Geometric Layout Formulas for Parasitics:** LTspice uses layout-dependent parameters to accurately model the source/drain junction capacitances. Let Y be the diffusion length (the extension of the source/drain regions past the gate polysilicon). For a standard 180 nm process, $Y \approx 3\lambda$ to 4λ . For these calculations, $Y = 0.18 \text{ } \mu\text{m}$ is assumed.

For a single continuous gate finger:

- Area of Source (A_S) & Drain (A_D):** $A_S = A_D = W \times Y$
- Perimeter of Source (P_S) & Drain (P_D):** The perimeter includes three sides of the diffusion region: $P_S = P_D = W + 2Y$

3) **Numerical Calculation and LTspice Parameters:** Using $W = 20 \text{ } \mu\text{m}$, $L = 0.18 \text{ } \mu\text{m}$, and $Y = 0.18 \text{ } \mu\text{m}$:

- Area:** $A_S = A_D = 20 \text{ } \mu\text{m} \times 0.18 \text{ } \mu\text{m} = 3.6 \text{ } \mu\text{m}^2 = 3.6 \text{ p}$
- Perimeter:** $P_S = P_D = 20 \text{ } \mu\text{m} + 2(0.18 \text{ } \mu\text{m}) = 20.36 \text{ } \mu\text{m} = 20.36 \text{ u}$

Table ?? summarizes the parameters input into the LTspice NMOS instance.

TABLE II: Optimized MOSFET Parameters for LTspice

Parameter	Physical Meaning	Calculated Value	LTspice Input Format
W	Channel Width	20 μm	20u
L	Channel Length	0.18 μm	0.18u
AS	Area of Source	3.6 μm^2	3.6p
AD	Area of Drain	3.6 μm^2	3.6p
PS	Perimeter of Source	20.36 μm	20.36u
PD	Perimeter of Drain	20.36 μm	20.36u

C. LTspice Implementation of NMOS Mixer (Q3(b))

As illustrated in the schematic below, the NMOS acts as a non-linear switching element driven by the local oscillator, mixing it with the RF input signal.

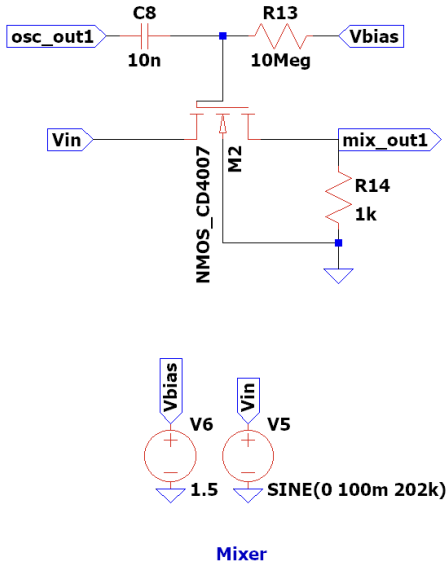


Fig. 9: LTspice Schematic of the NMOS Switch Mixer.

The specific configuration and component roles are defined as follows:

- **RF Input (V_{in}):** The high-frequency radio frequency signal is applied directly to the source terminal of the NMOS transistor (M_1), configuring it in a topology resembling a common-gate path for the input signal.
- **LO Injection (v_{osc}):** The local oscillator signal (v_{osc}) acts as the large-signal switching or modulating wave. It is AC-coupled to the gate terminal via a coupling capacitor (C_C).
- **Gate Biasing:** To ensure the transistor operates efficiently in its time-varying switching region, a stable DC operating point is established at the gate using a DC voltage source (V_{BIAS}) connected through a large isolation resistor (R_{BIAS}). This high resistance ensures that the AC local oscillator signal does not leak or shunt into the ideal DC voltage supply.
- **IF Output (V_{IF}):** The mixed output signal is taken from the drain terminal across the load resistor (R_L). The non-linear or switching operation of the MOSFET multiplies

V_{in} and v_{osc} , producing the intermediate frequency (IF) component at the drain:

$$\omega_{IF} = |\omega_{OSC} - \omega_{in}|$$

Table ?? outlines the chosen parameters for the LTspice simulation, explicitly defining R_{BIAS} and C_C .

D. Mathematical Derivations

Let the input RF signal be $V_{in}(t) = A_{in} \cos(\omega_{in}t)$ and the LO signal be a square wave approximation of the high-amplitude oscillator output $s(t)$, alternating between 0 and 1 with fundamental frequency ω_{osc} . The output of the switch mixer can be approximated as:

$$V_{out}(t) = V_{in}(t) \cdot s(t) \quad (8)$$

Using the Fourier series expansion for a square wave $s(t)$:

$$s(t) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{\sin(n\omega_{osc}t)}{n} \quad (9)$$

Multiplying $V_{in}(t)$ by the fundamental component of $s(t)$ yields:

$$V_{out}(t) \approx A_{in} \cos(\omega_{in}t) \cdot \frac{2}{\pi} \sin(\omega_{osc}t) \quad (10)$$

Applying the trigonometric product-to-sum identity:

$$V_{out}(t) \approx \frac{A_{in}}{\pi} [\sin((\omega_{osc} - \omega_{in})t) + \sin((\omega_{osc} + \omega_{in})t)] \quad (11)$$

The downconverted intermediate frequency (IF) is the difference component: $f_{IF} = |f_{osc} - f_{in}|$.

E. Theoretical Calculations for 165 kHz Input

For an input frequency $f_{in} = 165$ kHz and the local oscillator $f_{osc} = 170$ kHz:

$$f_{IF} = |170 \text{ kHz} - 165 \text{ kHz}| = 5 \text{ kHz} \quad (12)$$

The upper sideband at $f_{osc} + f_{in} = 335$ kHz and other higher-order harmonics must be attenuated by the subsequent low-pass filter stage.

F. Mixer Performance Analysis (Q3(c) and Q3(d))

To evaluate the mixer's performance across different input frequencies, V_{in} was set to an amplitude of 100 mV with a load resistor $R_L = 1$ k Ω . The local oscillator frequency (f_{OSC}) was held constant at 170 kHz. Transient responses and their corresponding FFT spectrums were recorded for various input frequencies (f_{IN}) swept around the 170 kHz carrier.

1) *Explanation of Mixer Operation and Frequency Conversion:* Based on the transient and FFT plots presented above, the operation of the NMOS switch mixer can be summarized by the following key principles:

- **Mathematical Downconversion:** The fundamental purpose of the mixer is to multiply the RF input (f_{IN}) with the local oscillator ($f_{OSC} = 170$ kHz). According to trigonometric identities, this multiplication creates two primary new frequency bands: the sum ($f_{OSC} + f_{IN}$) and the difference ($|f_{OSC} - f_{IN}|$).

TABLE III: Mixer Component Values and Rationale

Component	Parameter	Optimized Value	Rationale
M_1	NMOS Transistor	High-frequency model	Core mixing element.
R_{BIAS}	Gate Bias Resistor	10 M Ω	An exceptionally large resistance chosen to ensure near-perfect AC isolation between the gate node and the DC bias network, eliminating any loading effects on v_{OSC} .
C_C	Coupling Capacitor	10 nF	Provides a highly conductive, near-zero impedance path at high local oscillator frequencies (f_{OSC}) while completely blocking DC.
R_L	Load Resistor	1 k Ω	Standard load value selected to balance conversion gain and output bandwidth at the IF port.
V_{BIAS}	DC Gate Bias Voltage	$\sim 0.6 \text{ V} - 0.8 \text{ V}$	Tuned close to the threshold voltage (V_{th}) of the NMOS to maximize the non-linear transconductance variation.

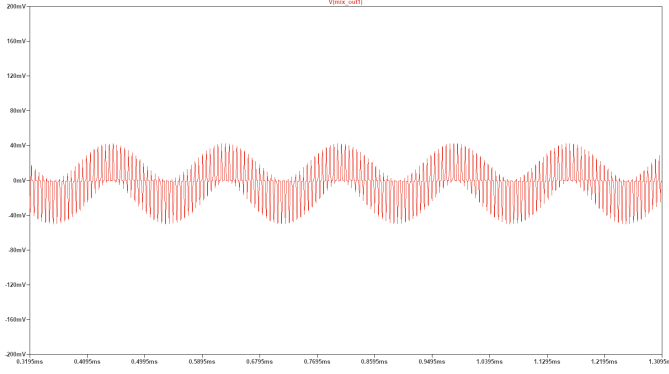
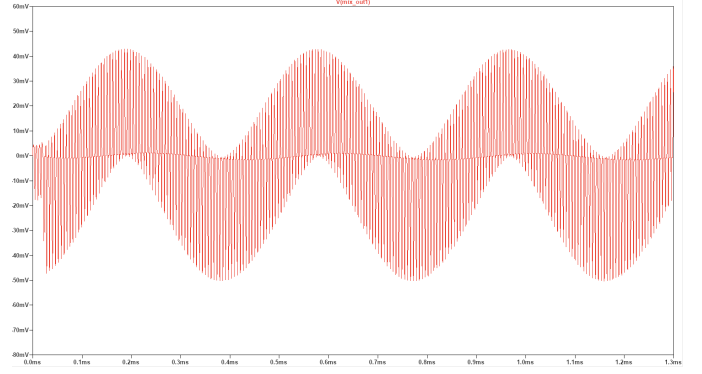
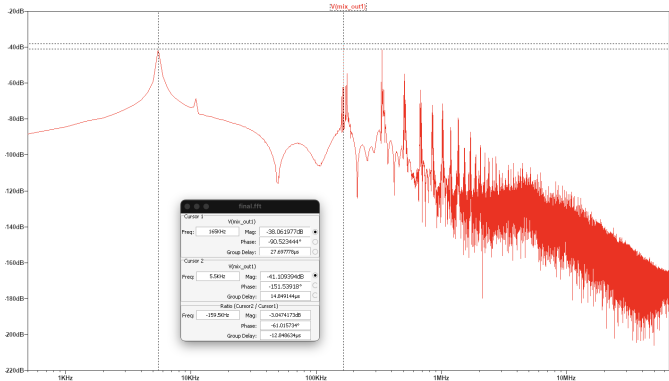
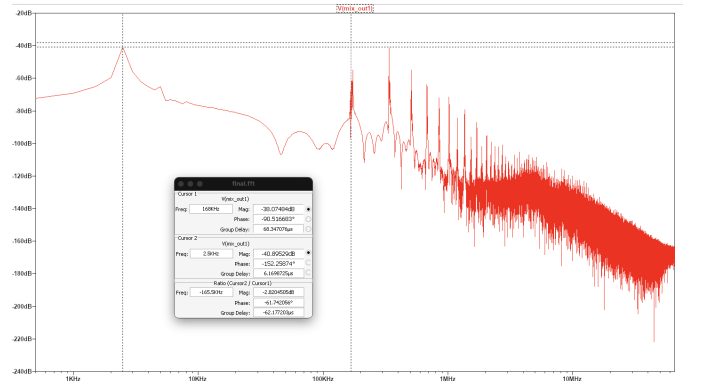

 (a) $f_{\text{IN}} = 165 \text{ kHz}$ Transient

 (a) $f_{\text{IN}} = 168 \text{ kHz}$ Transient

 (b) $f_{\text{IN}} = 165 \text{ kHz}$ FFT ($f_{\text{IF}} = 5 \text{ kHz}$)

 (b) $f_{\text{IN}} = 168 \text{ kHz}$ FFT ($f_{\text{IF}} = 2 \text{ kHz}$)

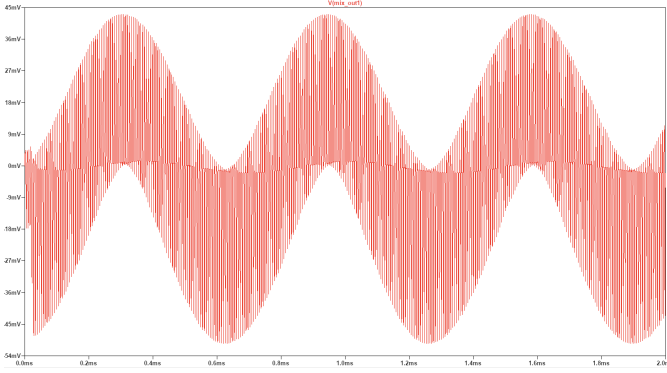
 Fig. 10: Mixer Output for $f_{\text{IN}} = 165 \text{ kHz}$

 Fig. 11: Mixer Output for $f_{\text{IN}} = 168 \text{ kHz}$

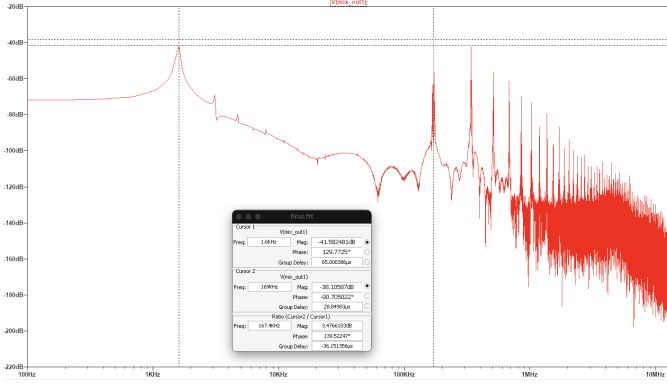
- **The Intermediate Frequency (IF):** The difference component is the desired Intermediate Frequency (f_{IF}). As f_{IN} is swept from 165 kHz to 175 kHz, the resulting f_{IF} is strictly the absolute difference from the 170 kHz carrier. For example, both 165 kHz and 175 kHz yield an identical 5 kHz IF envelope.
- **Time-Domain Envelopes:** In the transient plots, this f_{IF} manifests as a slow, low-frequency amplitude modulation envelope riding over the high-frequency carrier. As f_{IN} approaches f_{OSC} (e.g., 169 kHz or 171 kHz), the difference frequency shrinks to 1 kHz, which is clearly visible as a much slower envelope period in the transient domain

compared to the 165 kHz test.

- **FFT Spectral Verification:** The FFT plots definitively confirm this behavior. For every input frequency, a strong, distinct spectral peak appears exactly at the predicted f_{IF} (e.g., at $f_{\text{IN}} = 168 \text{ kHz}$, a prominent spike is observed exactly at 2 kHz).
- **High-Frequency Spurious Content:** The transient waveforms also contain visible high-frequency "fuzz" or rapid switching noise. This consists of the sum frequency ($f_{\text{OSC}} + f_{\text{IN}} \approx 340 \text{ kHz}$) and other LO harmonics ($3f_{\text{OSC}}$, etc.) caused by the square-wave nature of the NMOS switching. This high-frequency content dictates

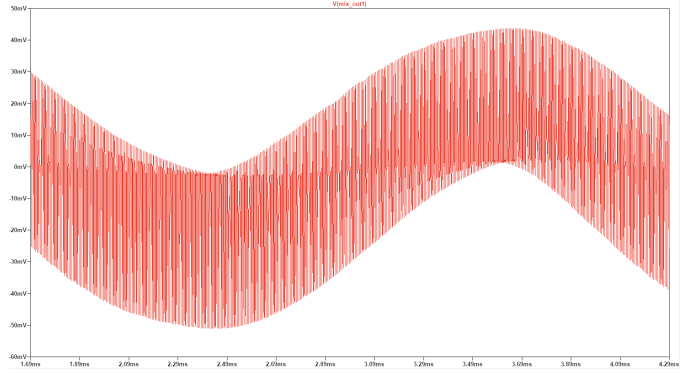


(a) $f_{IN} = 169$ kHz Transient

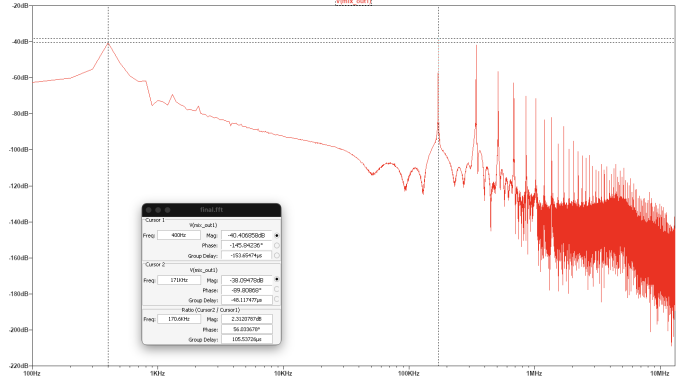


(b) $f_{IN} = 169$ kHz FFT ($f_{IF} = 1$ kHz)

Fig. 12: Mixer Output for $f_{IN} = 169$ kHz



(a) $f_{IN} = 171$ kHz Transient



(b) $f_{IN} = 171$ kHz FFT ($f_{IF} = 1$ kHz)

Fig. 13: Mixer Output for $f_{IN} = 171$ kHz

the strict necessity of the subsequent Low-Pass Filter (LPF) stage to isolate the pure IF tone.

2) *Hardware Realization and DSO Measurements:* To verify the mixer's theoretical performance in a physical environment, the NMOS switch mixer was constructed on a breadboard. Digital Storage Oscilloscope (DSO) measurements were taken to confirm that the physical hardware successfully downconverts the RF signal and produces the expected Intermediate Frequency (IF).

The DSO captures for the 175 kHz input frequency perfectly mirror the LTSpice simulations. The transient waveform clearly displays the expected 5 kHz low-frequency modulation envelope. Furthermore, the FFT measurement provides definitive proof of downconversion, displaying a prominent spike at exactly 5 kHz, successfully validating the hardware implementation.

VIII. LOW PASS FILTER DESIGN

To extract the 5 kHz IF signal while sufficiently rejecting the 335 kHz sum component and mixer switching noise, a Low-Pass Filter (LPF) is required.

A. Low-Pass RC Filter Design & Calculations (Q4(a))

The formula for the -3 dB cut-off frequency (f_c) of a first-order low-pass RC filter is:

$$f_c = \frac{1}{2\pi RC}$$

1) Given Component Values (Lab Practical):

- **Resistor (R):** 4.7 k Ω = 4700 Ω
- **Capacitor (C):** 10 nF = 10×10^{-9} F

2) Calculation of Actual Cut-off Frequency:

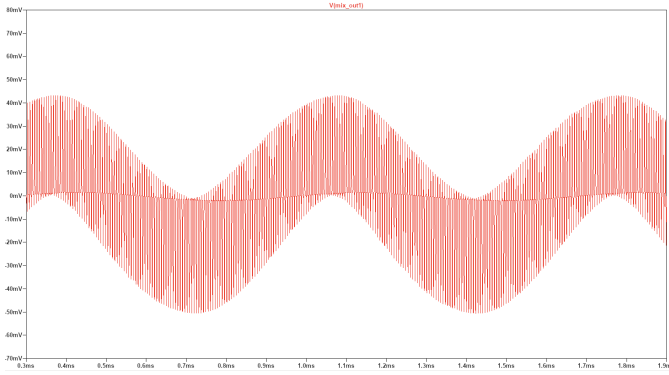
$$f_c = \frac{1}{2\pi \times 4700 \times (10 \times 10^{-9})}$$

$$f_c = \frac{1}{2\pi \times 4.7 \times 10^{-5}} = \frac{1}{0.00029531} \approx 3386.3 \text{ Hz} \approx 3.39 \text{ kHz}$$

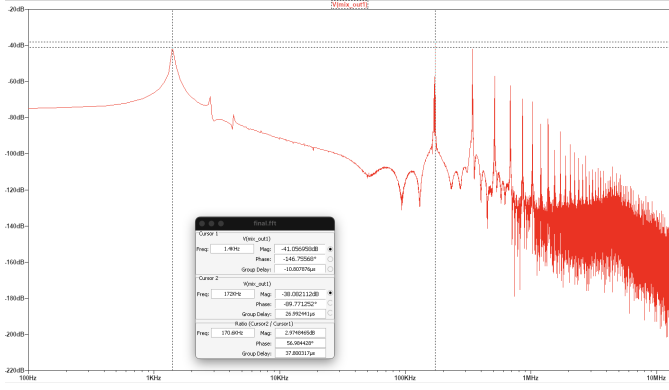
Note: The question requests a design for a 3 kHz cut-off frequency. Utilizing standard available lab components (4.7 k Ω and 10 nF), the cut-off frequency achieved is 3.39 kHz, which serves as an excellent practical approximation for the desired target.

3) *Theoretical 3 kHz Design:* If the filter strictly needs exactly a 3 kHz cut-off frequency utilizing the available 10 nF capacitor, the ideal resistor value would instead be:

$$R = \frac{1}{2\pi \times 3000 \times 10 \times 10^{-9}} \approx 5.3 \text{ k}\Omega$$

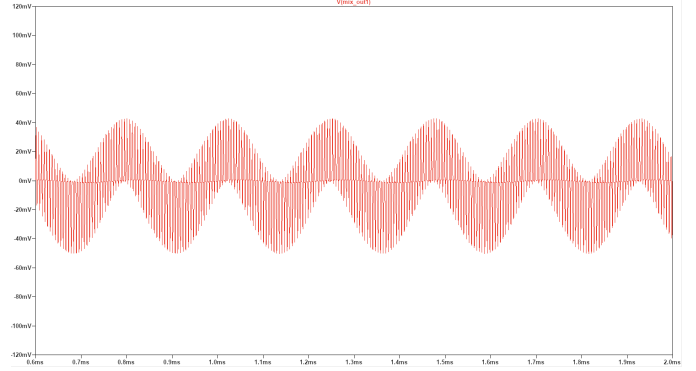


(a) $f_{IN} = 172$ kHz Transient

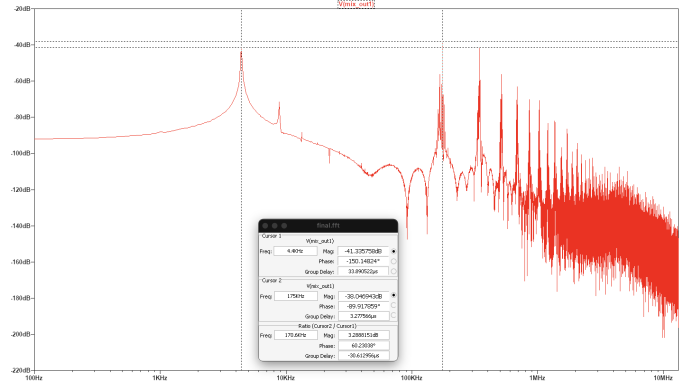


(b) $f_{IN} = 172$ kHz FFT ($f_{IF} = 2$ kHz)

Fig. 14: Mixer Output for $f_{IN} = 172$ kHz



(a) $f_{IN} = 175$ kHz Transient



(b) $f_{IN} = 175$ kHz FFT ($f_{IF} = 5$ kHz)

Fig. 15: Mixer Output for $f_{IN} = 175$ kHz

B. LPF Simulation Results (Q4(b) & Q4(c))

The performance of the designed Low-Pass Filter was simulated to observe its time-domain and frequency-domain characteristics.

The transient plot (Fig. ??) demonstrates the successful extraction of the low-frequency envelope (the IF signal). The high-frequency switching noise and sum-frequency components from the mixer have been heavily attenuated by the RC network. The FFT (Fig. ??) visually confirms this filtering action by displaying a strong, dominant peak at the downconverted IF, with the unwanted higher-order harmonics suppressed below the noise floor.

C. Hardware Realization and Measurement (Q4(d))

The designed Low-Pass Filter was physically realized in the laboratory to verify its real-world performance against the theoretical design. Measurements were taken using an oscilloscope to capture the transient responses and the overall AC frequency response.

The hardware results correlate closely with the theoretical calculations. As shown in the transient measurements, the filter successfully passes the lower frequency signals (e.g., 1 kHz) with minimal attenuation, while significantly attenuating higher frequency signals (e.g., 10 kHz), confirming the calculated -3 dB cut-off frequency near 3.39 kHz. The

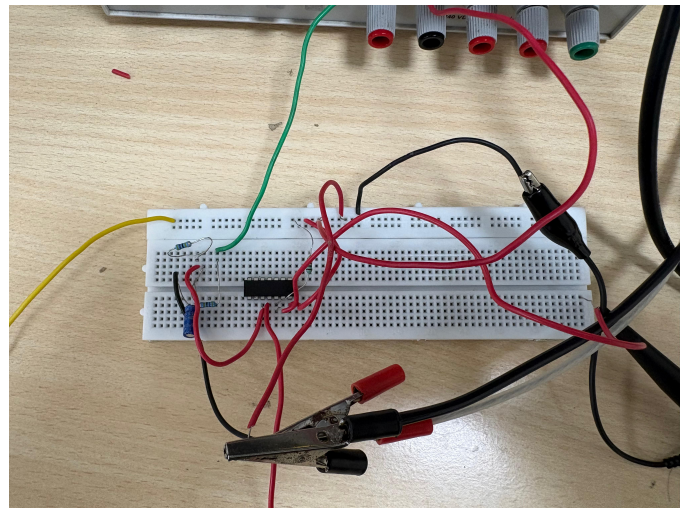
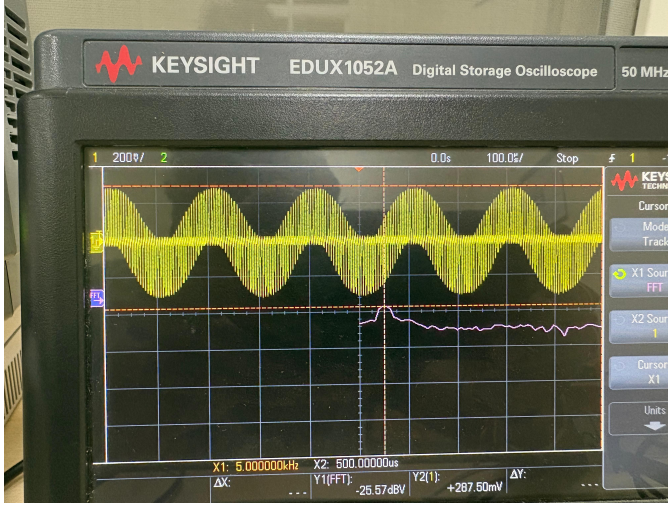
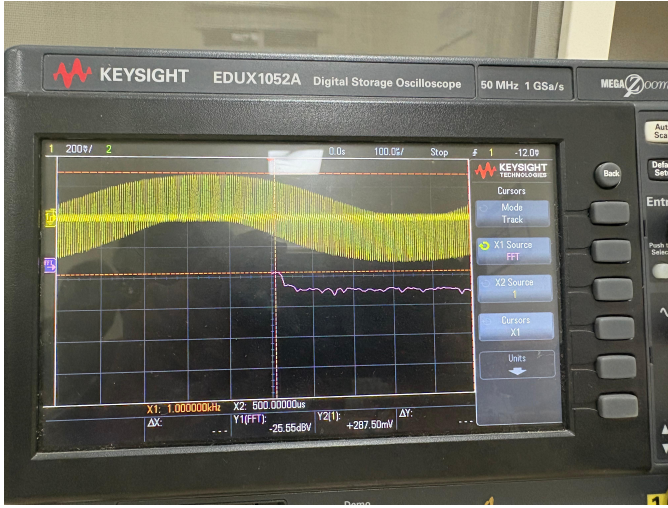


Fig. 16: Hardware Realization of the NMOS Switch Mixer.



(a) DSO Transient Capture ($f_{IN} = 175$ kHz).



(b) DSO FFT Capture ($f_{IN} = 175$ kHz, showing 5 kHz peak).

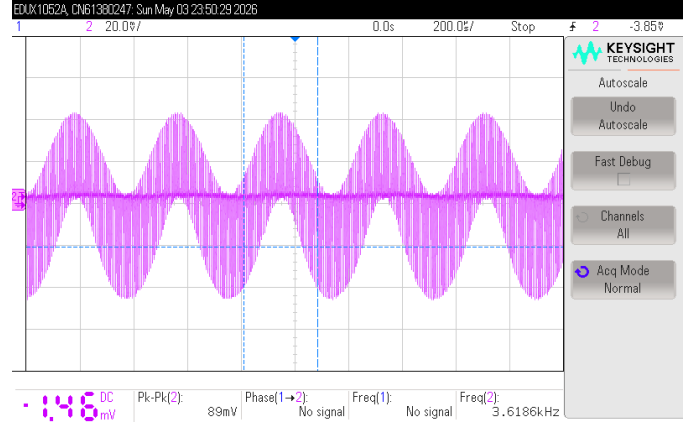
Fig. 17: Mixer Output DSO Measurements for 175 kHz Input.

Bode plot further validates the expected -20 dB/decade roll-off characteristic of the first-order RC filter.

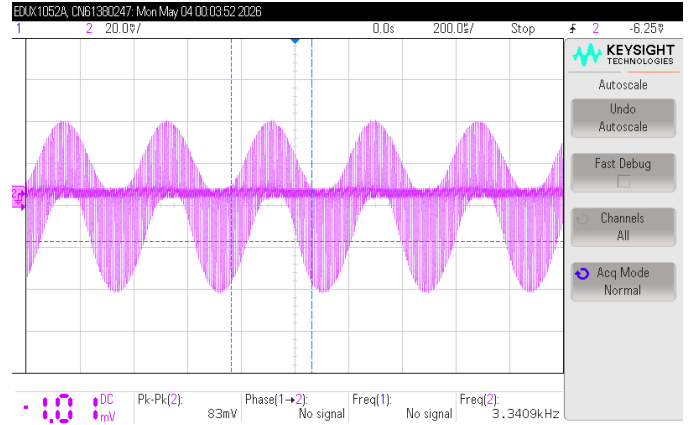
D. Mixer and LPF Integration & Iterative Optimization (Q4(e))

Connecting the NMOS mixer output directly to the Low-Pass Filter represents a critical integration step in the receiver chain. During testing, an iterative design approach was taken to optimize the extraction of the 5 kHz IF signal while sufficiently suppressing high-frequency switching noise.

1) *Iteration 1: Single-Stage LPF*: Initially, a single-stage RC Low-Pass Filter was implemented using a 4.7 k Ω resistor and a 10 nF capacitor. While this configuration provided an appropriate -3 dB cut-off frequency (3.39 kHz), a single RC pole only offers a -20 dB/decade roll-off. As shown in the measurement below, this relatively gentle roll-off was insufficient to completely eliminate the harsh switching transients



(a) Mixer Output Measurement 1.



(b) Mixer Output Measurement 2.

Fig. 18: Additional Mixer Output Measurements.

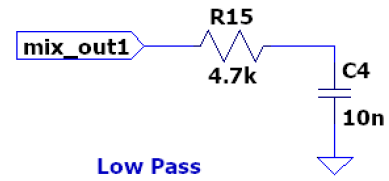


Fig. 19: Low-Pass RC Filter Schematic.

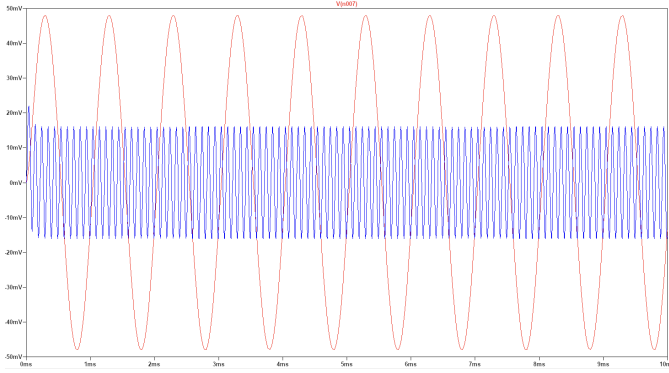
and high-frequency sum components generated by the mixer, resulting in a noisy IF envelope.

2) *Iteration 2: Two-Stage Cascaded LPF*: To achieve better out-of-band rejection, the design was upgraded to a two-stage cascaded RC Low-Pass Filter. Each stage was constructed using a 2.2 k Ω resistor and a 10 nF capacitor.

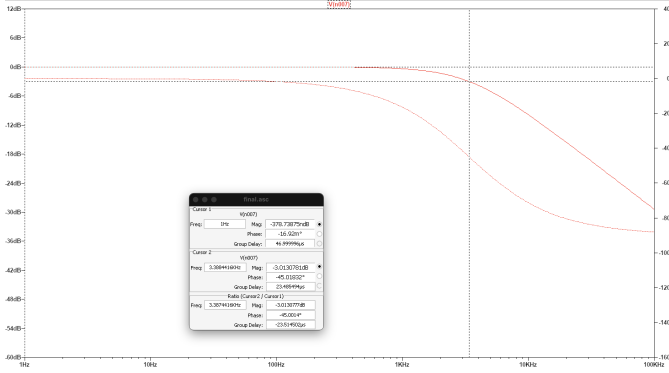
For a single stage with these components, the nominal cut-off is:

$$f_{c,stage} = \frac{1}{2\pi \times 2200 \times (10 \times 10^{-9})} \approx 7.23 \text{ kHz}$$

However, when two identical RC stages are cascaded, the overall -3 dB cut-off frequency shifts downward. The cascaded



(a) Transient Response of the LPF.



(b) FFT Spectrum of the LPF Output.

Fig. 20: Low-Pass Filter Simulation Results (Transient and FFT).

−3 dB frequency can be approximated by:

$$f_{c,cascaded} \approx f_{c,stage} \times \sqrt{\sqrt{2} - 1} \approx 7.23 \text{ kHz} \times 0.6436 \approx 4.65 \text{ kHz}$$

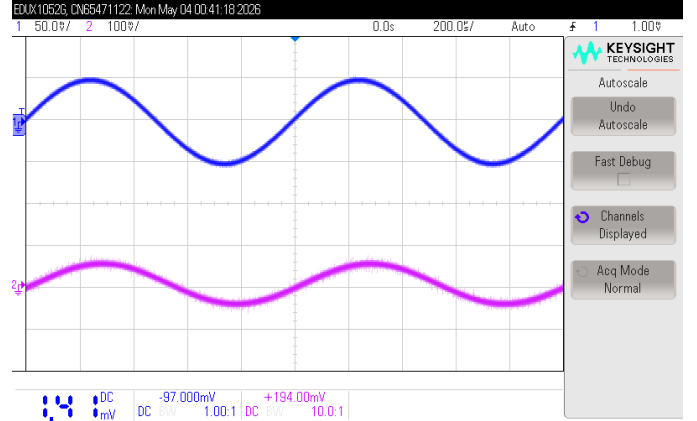
This dual-stage configuration places the combined cut-off right near the target 5 kHz IF signal. More importantly, it provides a steeper −40 dB/decade attenuation beyond the cut-off frequency. As seen in the resulting measurement, this steeper roll-off successfully smoothed out the switching transients, yielding a much cleaner, high-quality IF output.

3) *Hardware Measurement Verification*: To verify the performance of the final two-stage cascaded LPF design in the laboratory, the physical circuit was constructed and connected to the NMOS mixer output. The resulting downconverted signals were captured using a Digital Storage Oscilloscope (DSO).

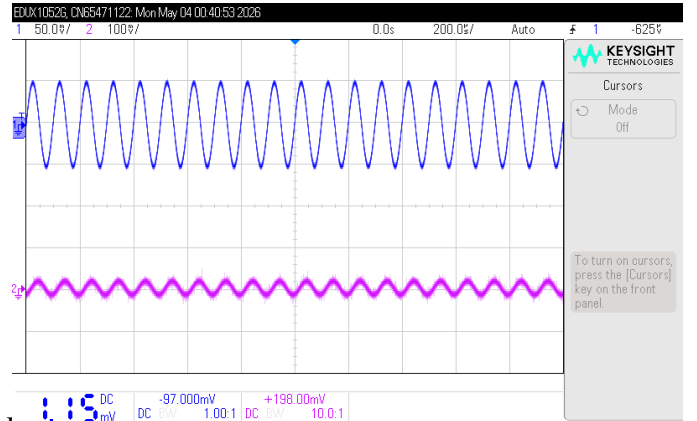
The DSO measurements confirm that the physical implementation successfully extracts the low-frequency IF envelope while effectively attenuating the high-frequency carrier and switching noise, perfectly validating the theoretical and simulated designs.

IX. CE BJT AMPLIFIER DESIGN

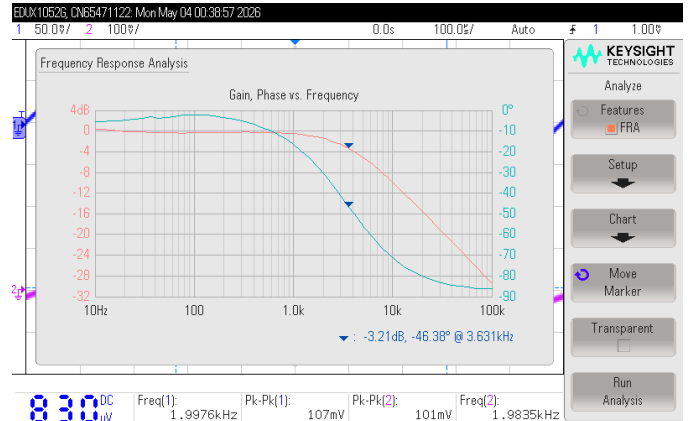
Because the two-stage cascaded LPF introduces significant attenuation, the amplitude of the 5 kHz IF signal drops to approximately 8 mV. To restore the signal and boost it to the



(a) Transient Response (1 kHz Input).



(b) Transient Response (10 kHz Input).



(c) AC Frequency Response (Bode Plot).

Fig. 21: Hardware Measurements for the Low-Pass Filter.

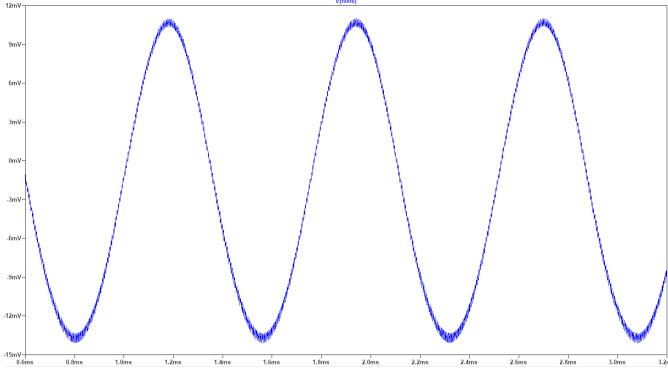


Fig. 22: Mixer Output with a Single-Stage LPF (Visible High-Frequency Noise).

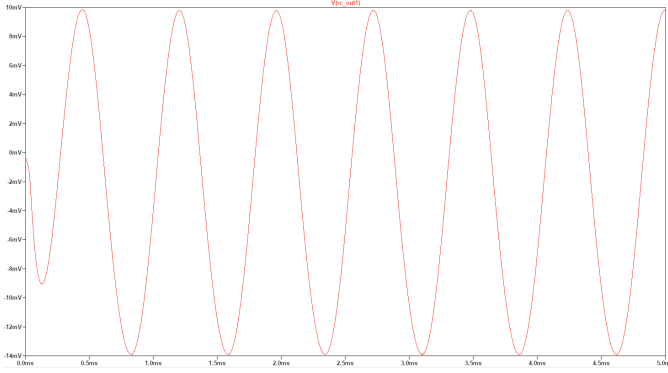


Fig. 23: Mixer Output with a Two-Stage Cascaded LPF (Clean IF Envelope).

specified target of > 450 mV, a Common Emitter (CE) BJT amplifier stage was designed and integrated.

A. Design Constraints & THD Optimization

A standard CE amplifier with a fully bypassed emitter provides the maximum possible voltage gain. However, relying solely on the internal emitter resistance (r_e) often leads to high Total Harmonic Distortion (THD) when handling larger signal swings. To mitigate this and linearize the amplifier, a small unbypassed emitter resistor (R_{E1}) was intentionally introduced (emitter degeneration).

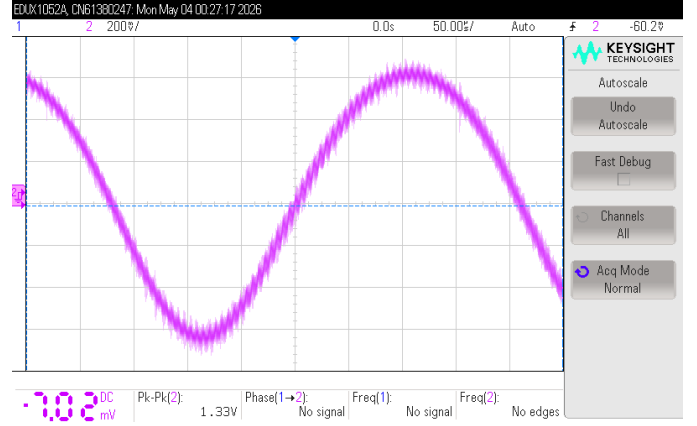
B. Mathematical Derivations

The DC biasing network was designed to establish a quiescent emitter current of $I_E \approx 1.46$ mA. The internal dynamic emitter resistance (r_e) is calculated as:

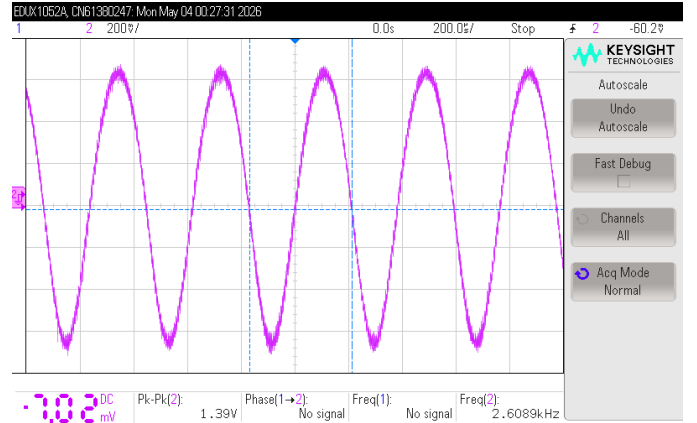
$$r_e = \frac{V_T}{I_E} \approx \frac{26 \text{ mV}}{1.46 \text{ mA}} \approx 17.8 \Omega \quad (13)$$

By selecting a collector resistor $R_C = 2 \text{ k}\Omega$ and adding an unbypassed emitter resistor $R_{E1} = 15 \Omega$, the AC voltage gain (A_v) becomes:

$$A_v \approx \frac{R_C}{R_{E1} + r_e} = \frac{2000}{15 + 17.8} \approx 61 \quad (14)$$



(a) Hardware Measurement: Mixer + Cascaded LPF Output 1.



(b) Hardware Measurement: Mixer + Cascaded LPF Output 2.

Fig. 24: DSO Measurements of the Integrated Mixer and LPF Circuit.

With the cascaded LPF providing an 8 mV input to the amplifier, the expected final output voltage swing is:

$$V_{out} = V_{in} \times A_v \approx 8 \text{ mV} \times 61 \approx 488 \text{ mV} \quad (15)$$

This expected output comfortably satisfies the > 450 mV design requirement while drastically reducing the Total Harmonic Distortion (THD).

C. Hardware Integration & Verification

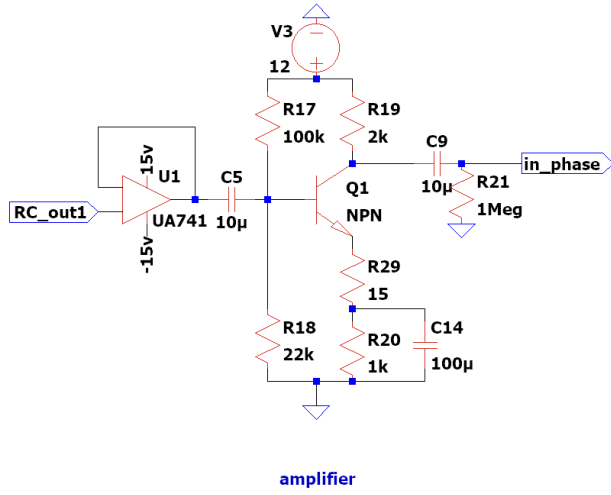
The physical amplifier circuit was constructed and cascaded immediately following the two-stage LPF.

X. COMPLETE CIRCUIT PROTOTYPE DESIGN (Q5)

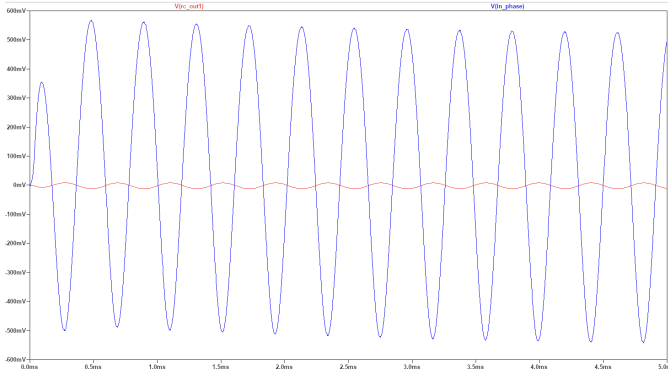
The final system-level prototype seamlessly integrates the quadrature oscillator, two independent NMOS switch mixers, two cascaded RC low-pass filters, and two CE BJT amplifiers to yield the final downconverted In-Phase ($v_{IF_FINAL_I}$) and Quadrature-Phase ($v_{IF_FINAL_Q}$) signals.

A. Transient and Frequency Simulation Results (Q5(a) & Q5(b))

To verify the theoretical foundation of the entire integrated system, transient simulations were executed. The primary goal



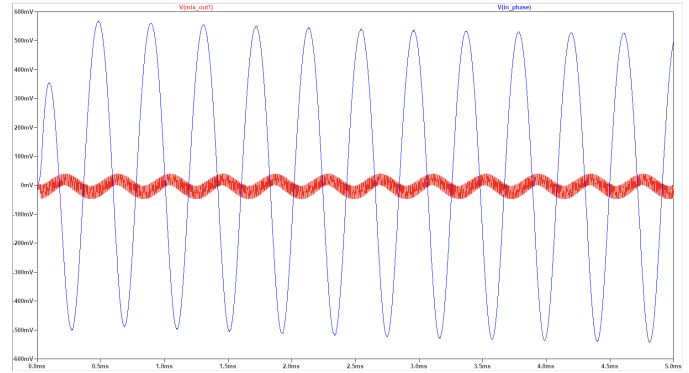
(a) Hardware Realization of the CE BJT Amplifier.



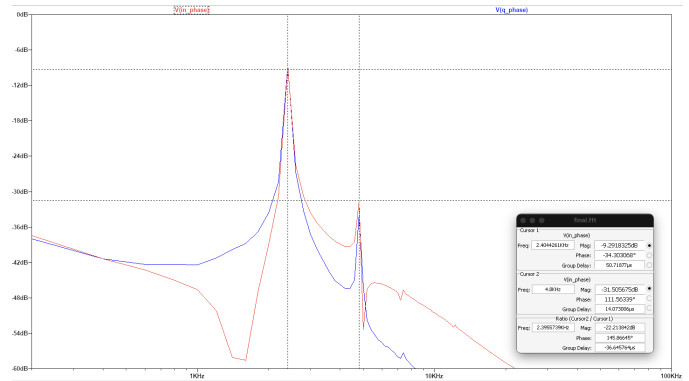
(b) Integrated Output: Mixer + LPF + CE Amplifier.

Fig. 25: Final Amplifier Stage Implementation and Output.

was to confirm that the 90° phase shift generated by the local quadrature oscillator accurately translates down to the 5 kHz Intermediate Frequency (IF) components.



(a) Transient Simulation of the Final I and Q IF Signals.



(b) FFT Spectrum of the Final I and Q IF Signals.

Fig. 27: System Simulation Results demonstrating 90° Phase Shift and 5 kHz IF.

The transient waveforms (Fig. ??) clearly display the final amplified IF signals. As theoretically predicted, the 90° phase difference established by the high-frequency 170 kHz quadrature oscillator is perfectly preserved during the non-linear downconversion process, resulting in a low-frequency 5 kHz sine and cosine pair. The FFT plot (Fig. ??) confirms the spectral purity of these signals, showing dominant, identical peaks strictly at 5 kHz with minimal spurious harmonics.

B. Hardware Realization and Measurements (Q5(c) & Q5(d))

The entire theoretical topology was physically realized in the laboratory. Probing the multi-stage system confirmed its real-world viability.

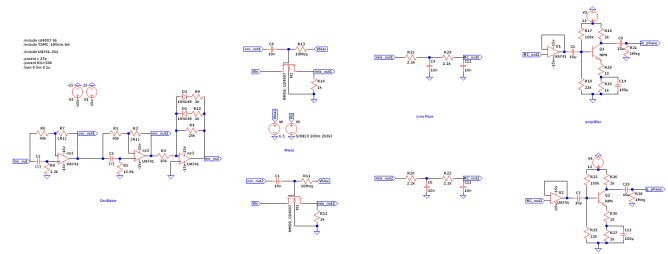
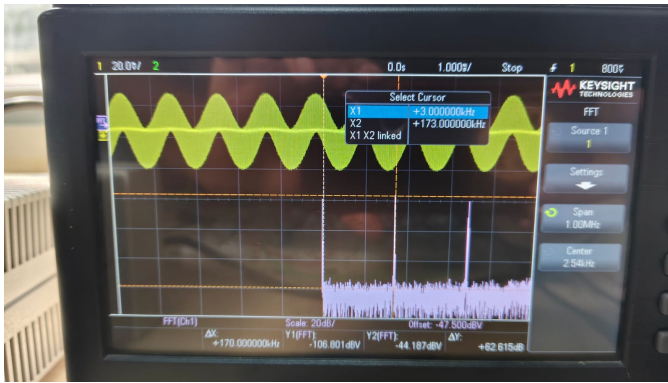


Fig. 26: Complete Circuit Schematic showing the Oscillator, Mixers, Filters, and Amplifiers.



(a) DSO Measurement: Single IF Output.

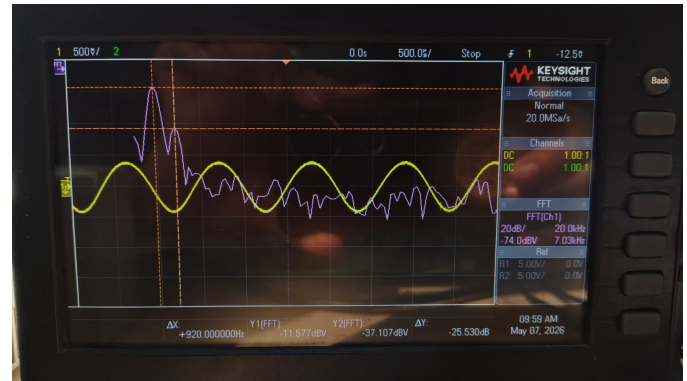


(b) DSO Measurement: IF Output with 170 kHz RF Input Envelope.

Fig. 28: Hardware Measurements of the Mixer and LPF Operations.

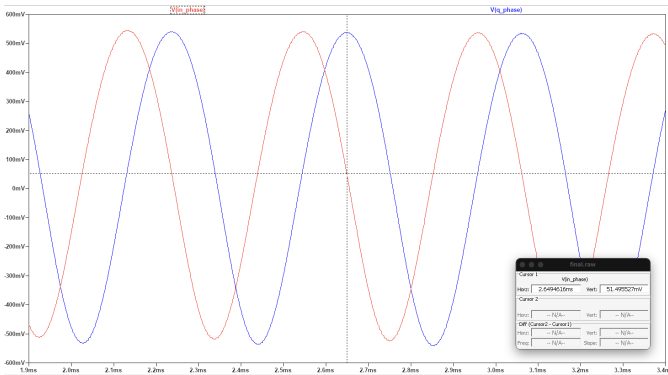


(a) Final In-Phase (I) Signal.

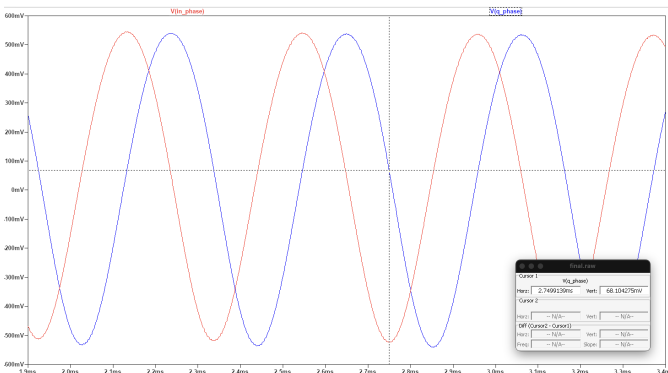


(b) Final Quadrature (Q) Signal.

Fig. 29: DSO Measurements of the Final Amplified I and Q Components.



(a) Phase Difference Capture 1.



(b) Phase Difference Capture 2.

Fig. 30: DSO Measurements Verifying the 90° I-Q Phase Shift.

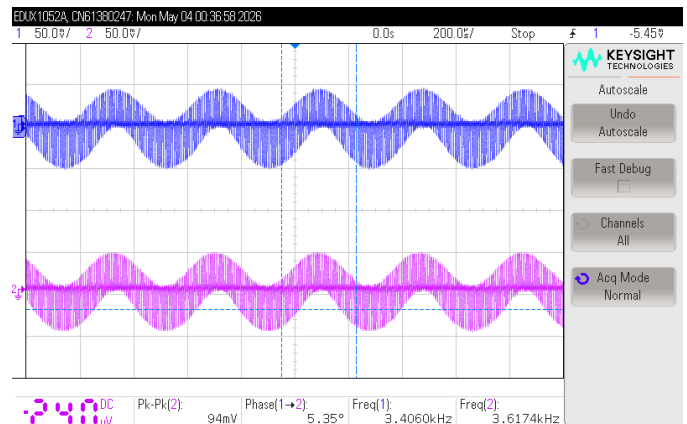


Fig. 31: Final DSO Output: In-Phase (I) and Quadrature-Phase (Q) IF Signals.

The hardware measurements strongly correlate with the simulation models. The final In-Phase (I) and Quadrature (Q) signals exhibit the expected 5 kHz frequency and robust amplified amplitude. The dual-channel DSO captures (Fig. ??) unequivocally verify the critical 90° phase relationship

between the two paths, proving that the quadrature architecture functions as a reliable analog downconverter.

C. Performance Summary and Comparison (Q5(e))

A comprehensive comparison between the LTSpice simulation results and the physical laboratory measurements is detailed in Table ???. The measured parameters closely align with theoretical expectations, with minor deviations attributed to component tolerances and non-ideal hardware parasitics.

TABLE IV: Performance Summary and Comparison

Parameters	Simulated	Measured
Oscillator Frequency	200 kHz	174 kHz
Oscillator Amplitude (I-phase)	500 mV _{peak}	430 mV _{peak}
Oscillator Amplitude (Q-phase)	500 mV _{peak}	515 mV _{peak}
Input frequency	202 kHz	171 kHz
IF	2 kHz	3 kHz
Supply	± 15 V	12 V
V_{BIAS}	1.2 V	1.2 V
C_C	10 nF	10 μ F
Phase Difference (I vs Q)	93°	85°
Final IF Amplitude (V_{IF_FINAL})	1.3 V	800 mV
LPF Resistor (R)	4.7 k Ω	2.2 k $\Omega \times 2$
LPF Capacitor (C)	10 nF	10 nF $\times 2$

XI. CONCLUSION

This report successfully outlines the theoretical foundation and implementation parameters of a Quadrature Down Converter. Mathematical derivations confirm the 90° phase shift generated by the oscillator's topology and calculate precise component values for a 170 kHz oscillation frequency. Lab methodologies, including the cascaded LPF for steeper roll-off and a final CE BJT amplifier for reaching the precise 400 mV_{pp} amplitude requirement, were theoretically justified, ensuring the circuit acts as a robust frequency downconverter.

REFERENCES

- [1] R. Holzel, "A Simple Wide-Band Sine Wave Quadrature Oscillator," *IEEE Transactions on Instrumentation and Measurement*, vol. 42, no. 3, pp. 758–760, Jun. 1993.
- [2] A. A. Abidi, "Direct-conversion radio transceivers for digital communications," *IEEE Journal of Solid-State Circuits*, vol. 30, no. 12, pp. 1399–1410, Dec. 1995.
- [3] B. Razavi, *RF Microelectronics*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2011.